Benjamin van Duin

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Education

University of North Carolina at Chapel Hill | *GPA: 3.95* August 2023 – May 2027
B.S. in Environmental Science (Ecology & Natural Resources) & Geospatial Data Science *Chapel Hill, NC*

Experience

Undergraduate Research Assistant April 2025 – Present
University of North Carolina Wildlife Biogeography Lab *Chapel Hill, NC*

- Classified thousands of mammal photographs by species, using Digikam to update image metadata.
- Conducted fieldwork across approximately 10 trips to Albemarle Sound research sites to deploy, move, and collect bucket and standing camera traps.

Resident Advisor Mentor August 2026 – Present
Carolina Housing *Chapel Hill, NC*

- Supervised and mentored a team of 27 Resident Advisors through daily office hours, while continuing full RA duties, including safety inspections, community programming, and conflict mediation.
- Reviewed, approved, and refined RA program proposals by giving feedback to improve event quality.
- Managed a $14,640 RA programming budget and a $1,215 staff development budget on a P-card, ensuring every expenditure complied with university procurement policy.

Resident Advisor August 2024 – May 2026
Carolina Housing *Chapel Hill, NC*

- Monitored the safety of approximately 300 residents in the Manning West Community by conducting regular inspections and responding to emergencies.
- Built community among residents by designing and hosting 2–3 programs on connection, resilience, and inclusion monthly.
- Mediated roommate conflicts and guided approximately 30 residents through collaborative living agreements.

FANgineer/FANgineer Leader August 2022 – November 2024
REVELxp (Carolina Athletic Hospitality) *Chapel Hill, NC*

- Led a FANgineer team enforcing premium seating access at Kenan Stadium, the Dean Smith Center, and Boshamer Stadium.

Projects

Coastline Change Analysis | *Google Earth Engine, QGIS* January – May 2025

- Quantified 36 years (1988–2024) of coastline change in the Mekong Delta, Vietnam, using Landsat 5/8 remote sensing imagery and unsupervised clustering in Google Earth Engine, finding a net land loss of ~78 km² (350 km² erosion vs. 272 km² accretion) similar to published estimates.
- Documented findings in a technical report linking observed erosion to a measured decline in suspended sediment concentration, supporting hypothesized dam-driven disruption of delta sediment supply.

Species Distribution Model | *R/RStudio* January – May 2025

- Built a species distribution model in R using GBIF presence data and spatial temperature and precipitation covariates, fitting a binomial logistic regression (AUC = 0.71) to predict and map habitat suitability.
- Validated predictions through spatial analysis against a decade of eBird abundance data, finding suitability correlated with temperature change (r = 0.50) but not with observed abundance trends.

Population Exposure to PM2.5 in Europe | *R/RStudio* January – May 2026

- Mapped PM2.5 exposure across Europe, finding 28.7 million people above the WHO guideline, by combining EEA and WorldPop data using area-weighted spatial analysis in R.
- Found population density explains under 1% of variance in PM2.5 (R^2 = 0.005) despite strong spatial clustering (Moran's I = 0.86), by running OLS and spatial error regression with LISA clustering and documenting the results in a technical report.

Licenses & Certificates

OSHA 40-Hour HAZWOPER Training (Safety Unlimited) October 2026
Remote Pilot Certificate (FAA Part 107) October 2026

Skills

GIS & Remote Sensing: GIS, ArcGIS, QGIS, Google Earth Engine, Landsat, spatial analysis, spatial statistics
Analysis & Field: R/RStudio, data analysis, mapping, fieldwork, technical report writing, Microsoft Office Suite
Languages: English (native), Dutch (native)